

Synthesis and Characterization of Nickel Ferrite Nanoparticles

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A dissertation submitted in fulfilment
of the requirement for the degree of

Bachelor of Physics



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Synthesis and Characterization of Nickel Ferrite Nanoparticles

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CERTIFICATE

This is to certify that the thesis entitled **Synthesis and Characterisation of Nickel Ferrite Nanoparticles** is an authentic record of the research work carried out by **Ms. Haritha P H** in partial fulfillment of the requirements for the award of degree of Bachelor of Science in Physics (2020-23) under the University of Calicut. The work presented in this thesis has not been submitted for any other degree or diploma earlier.

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CERTIFICATE (Supervisor)

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Palakkad

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Akshaya B, Varsha S, Nixon A

OBJECTIVE

The following are the objectives of our work:

1. Synthesis of NiFe_2O_4 ferrite nanoparticles using sol-gel auto combustion method and co-precipitation method
2. X-ray diffraction (XRD) studies of the synthesized NiFe_2O_4 ferrite nanoparticles.
3. Characterization of synthesized NiFe_2O_4 by UV-Visible Spectroscopy.
4. Characterization of synthesized NiFe_2O_4 by Fourier Transform Infrared Spectroscopy.

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Chapter 1

INTRODUCTION

1.1 Introduction to Nanotechnology

The prefix nano in the word nanotechnology means a billionth (1×10^{-9}). Nanotechnology deals with various structures of matter having dimensions of the order of a billionth of a meter. In recent years nanotechnology has become one of the most important and exciting forefront fields in Physics, Chemistry, Engineering and Biology. It shows great promise for providing us in the near future with many breakthroughs that will change the direction of technological advances in a wide range of applications.

Nanotechnology includes the production, characterization, and applications of materials that have dimensions smaller than 100 nm in atleast one direction. The principal parameters of nanoparticles are their shape, size, surface characteristics and inner structure. [2]

Nanoscale functional units are involved in some way or another in most physical, chemical, or biological processes, from catalysis to human metabolism to field-effect transistors. They have unusual chemical, physical, and biological properties that are not found in ordinary bulk materials. Thus, the methods and results of nanotechnology alter the profile of practically all the traditional natural and engineering sciences. Because of the very wide-ranging scope of research in nanotechnology, its fast-moving nature, and the impossibility of precisely defining the boundaries of the subject, a really exhaustive presentation of the field is practically unachievable.

There are a lot of advantages and some disadvantages about nanotechnology.

1.1.1 Advantages of nanotechnology

Nanotechnology offers ideas and solutions for many fields of activity, such as medicine, industry, information, communication or agriculture. [4]

- Basic medical applications for nanotechnology are: diagnosis, drug delivery and tissue engineering.
- Nanotechnology made revolutions in the field of manufacturing. For example, nanotubes, nanoparticles, aerogels, nanobots and nano factories could help construct novel materials and objects.
- Nanotechnology may transform the ways in which we obtain and use energy solar panels can be constructed more economically, leading to more efficient energy storage devices.
- The field of electronics is set to be revolutionized by nanotechnology. Quantum dots, for example, are tiny light-producing cells that could be used for illumination or for purposes such as display screens.

1.1.2 Disadvantages of nanotechnology

Along with the positive changes brought by nanotechnologies, they may also cause toxic effects for human health and for environmental security. The toxic effects have been recognized for pulmonary, reproductive, cardiac, digestive, cutaneous and immune levels. Nanomaterials can also cause air, water and most importantly a soil persistent form of pollution, which is too small to be detected easily, making nano-pollution another man-made unwanted environmental impact, with uncertain effects in the long term. There is also a possibility for loss of jobs in traditional farming and manufacturing industries. Manufacturing of nanomaterials is difficult, which in turn increases the expense of products made by nanotechnology. [4]

1.2 Nanomaterials and its importance

In recent years, considerable progress in synthesizing one-dimensional (1D) magnetic nanostructures, such as nanofibers, nanotubes, and nanoribbons, has been inspiring many researchers to investigate their magnetic-domain configurations and to develop their electromagnetic nanoscale device applications. 1D magnetic nanostructures possess remarkable

geometrical limitations being comparable to the critical magnetic lengths such as exchange length and domain wall width and can effectively overcome the serious aggregation of nanoparticles. Coaxial nanostructures exhibit additional remarkable effects and have potential applicability as multifunctional materials. Bulk materials are widely used in electro-acoustic devices, motors, electricity generators, etc. On the other hand, thin films of ferrites are widely applicable in the fabrication of magnetic and magneto-optic devices. In the thin film form, ferrites were proposed to be the important materials for nonreciprocal microwave devices because of its high electrical resistivity and frequency dependent permittivity. Precise control over morphologies of bulk/thin film materials is an important aspect from the point of view of their applications and also important in material science research. [1]

1.3 Ferrites and their significance:

Microcrystalline ferrites form the basis of materials currently used for magnetic information recording and storage. Nanocrystalline ferrites and magnetic carriers based on them will increase the recorded information density. The general chemical formula of ferrites possessing the structure of the mineral spinel MgAl_2O_4 is MFe_2O_4 , where M represents a divalent metal ion with an ionic radius approximately between 0.6 and 1 Å. In the case of simple ferrites, M is one of the divalent ions of the transition elements Mn, Fe, Co, Ni, Cu and Zn, or Mg and Cd [4]. The presence of Fe^{2+} , Ni^{2+} , Co^{2+} and Mn^{2+} can provide unpaired electron spins and therefore contribute to the magnetic moment of spinel.

1.3.1 Crystal structure of spinal ferrite

Ferrites are mixed metal oxides with iron (III) oxides as their main component. Ferrites Crystallize in three crystal types: [13]

1. Spinel type with cubic crystal structure and general formula $\text{M}^{2+}\text{Fe}_2\text{O}_4$ where M = Fe, Mn, Mg, Ni, Zn, Cd, Co, Cu, Al or a mixture of these.
2. Garnet type with cubic crystal structure and general formula $\text{Ln}^{3+}\text{Fe}_3\text{O}_{12}$ where Ln = Y, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, or Lu.

3. Magnetoplumbite type with hexagonal crystal structure and general formula $M^{2+} Fe_{12} O_{19}$ where $M = Ba$ or Sr .

Spinel ferrites are the simplest of these three groups. A detailed picture of the spinel ferrites is shown below. Spinel ferrites crystallize into the spinel structure, which is named after the mineral spinel, $MgAl_2O_4$ (Fig. 1.3.1). This structure was first determined by Bragg and Nishikawa. Spinel structure is determined primarily by the oxygen ion lattice. The oxygen anions are packed in a face centered cubic arrangement such that there are two kinds of interstitial space between the anions-tetrahedrally co-ordinated A sites and octahedrally co-ordinated B sites. There are eight formula units of MFe_2O_4 in a unit cell. Thus in a unit cell of 32 oxygen anions, there are 64 tetrahedral sites and 32 octahedral sites. It turns out that of the 64 tetrahedral sites only 8 are occupied and out of 32 octahedral sites, only 16 are occupied.

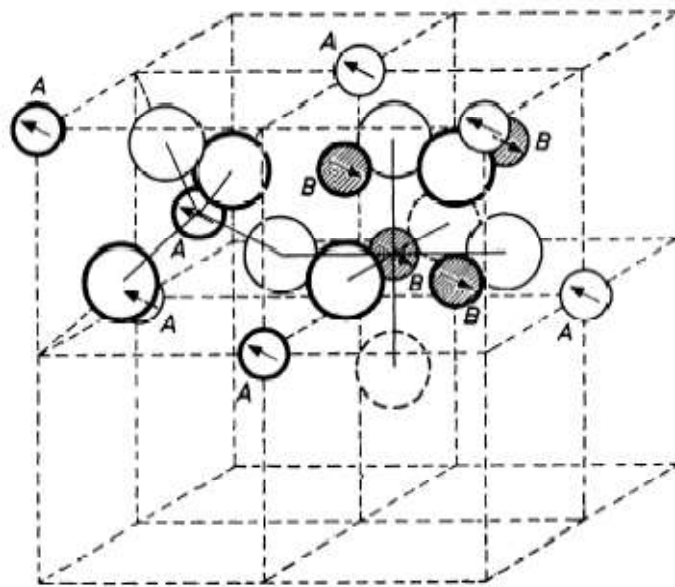


Figure 1.3.1 Unit cell of spinel structure

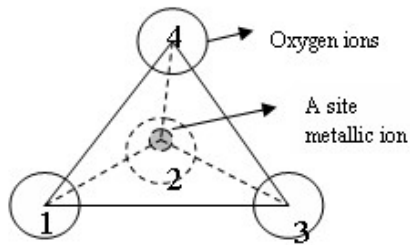


Figure 1.3.2 Tetrahedral Coordination

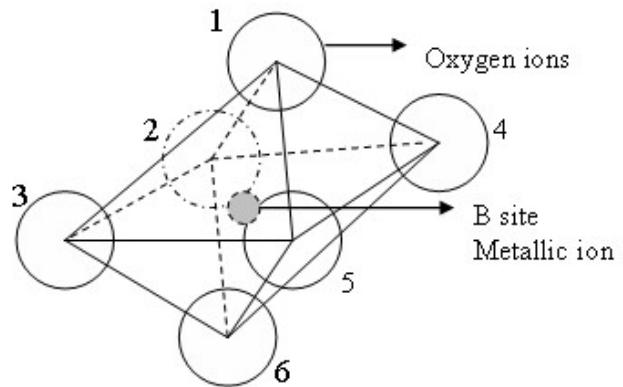


Figure 1.3.3 Octahedral Coordination

Picture Courtesy: [4]

1.3.2 Ferrimagnetic materials

Ferrimagnetism is a type of magnetic ordering. In ferrimagnets the moments are in an anti-parallel alignment, but they do not cancel. The best example of a ferrimagnetic mineral is magnetite (Fe_3O_4). Two iron ions are trivalent, while one is divalent. The two trivalent ions align with opposite moments and cancel one another, so the net moment arises from the divalent iron ion. Only the iron ions are magnetic. Ferrites are oxides of iron with the formula MFe_2O_4 , where M is a divalent ion such as nickel, zinc, cadmium, manganese, or magnesium. The ferric iron ion is trivalent, and the oxygen ion accepts two electrons. In general, M can also be a divalent iron, forming magnetite (Fe_3O_4). The crystal structure is called spinel. Ferrites are electrical insulators with magnetic ordering. Their insulating quality makes them useful as magnetic cores. [14]

1.3.3 Applications of Ferrites

Ferrites are ferrimagnetic materials, a broad class of magnetic oxides with its remarkable structural, electrical, dielectric and magnetic properties making them suitable for a variety of applications. These properties strongly depend upon the method of preparation. Several decades ago, research efforts were focused on active solids, which are of great scientific and technological importance. The exceptional properties of ferrites include significantly high saturation magnetization, a high electrical dc resistivity, considerably low electrical losses, with a high chemical stability. They are extensively used in various applications starting from simple lifting magnets to the most complex microwave communication system employed in outer space.

They can be used in satellite communication, radar, memory of computer, radio, television, videotape, etc. The most efficient power supplies used in electronic appliances are the switch mode power supplies (SMPS), which have ferrite as a major part. Magnetic resonance imaging (MRI) enhancement, catalysis, sensors, and pigments are the few of the developing areas of the applications of ferrites. These novel applications of the ferrites are possible due to the synthesis of the ferrites at nanoscale [1]. Thus, nanoscale size of the spinel ferrites has opened up the door for intensive research to utilize their properties for different novel applications. It includes applications of the ferrites as the material for magnetocaloric effect (MCE), sensors, biosensors, supercapacitors, etc.

In this work, NiFe_2O_4 (Nickel Ferrite) prepared through co-precipitation method and sol-gel method are analyzed, compared and studied through various characterisation techniques including X-ray Diffraction, UV-Visible Spectroscopy and Fourier Transform Infrared Spectroscopy.

Chapter 2

SYNTHESIS AND CHARACTERISATION METHODS

2.1 Introduction on Nanomaterial Synthesis

Ferrites, the magnetic ceramics consisting of Fe^{3+} as the major constituent, were initially prepared by a solid-state reaction mechanism known as standard ceramic method. This synthesis method involves stoichiometry mixing and grinding (for long hours) powders of the constituent oxides chemicals, carbonates, and heating them at relatively high temperature (up to $\sim 1000^\circ\text{C}$) with intermediate grinding, if necessary. The ceramic method is thus called as a time-consuming and a brute force method. It also yields particles of comparatively large course because of high temperature treatment. One of the other major drawbacks is the irregularity in shape, size, and stoichiometry, which may result in the non reproducible ferrite products. To overcome these drawbacks of standard ceramic method, numerous chemical methods were explored including *wet chemical coprecipitation method*, *sol-gel methods*, thermal decomposition method, micelle and reverse micelle method, Pechini method, etc.

Chemical synthesis approaches have the advantage over the ceramic method because:

- i. it requires low temperatures,
- ii. it yields fine particles of nanosize,
- iii. the method is easy and cheap and requires less time,
- iv. stoichiometry does not change,
- v. the product is reproducible.

Chemical method essentially makes use of simple reactions such as inter collation, ion-exchange, hydrolysis, dehydration, and reduction that can be carried out at relatively low processing temperature. Developing a method of synthesis of ferrites with optimum

properties is difficult and complex as well. The main problem is induced since most of the properties requirement for ferrite applications are extrinsic and not intrinsic, where the ferrite is defined by its chemistry and crystal structure, as well as requires knowledge and control of different parameter of its microstructures, such as bulk density, grain/crystallite size, percentage porosity, and their intra- and inter-granular distribution. The chemical synthesis methods cited above are few among the methods suitable in these criteria. In the area of magnetism of fine particles, modifications in saturation magnetization and magnetic anisotropy have been affected by the reduction of particle and grain sizes. [1]

Generally, two main approaches are adapted for the synthesis of nanoparticles; they are: [3]

2.1.1 Bottom-up Method

Bottom-up approach is dealing with the projection and development of new materials or devices, so that the produced nanostructures are assembled and organized into higher macro scale structures by principles of molecular recognition. Examples of bottom-up technique are self-assembly of nanomaterials, sol-gel technology, electrodeposition, physical and chemical vapour deposition (PVD, CVD), Epitaxial growth etc.

2.1.2 Top-down Method

In the top-down approach the existing materials are miniaturized into the nanoscale, and then the molecules rearranged to achieve the desired properties. Top-down technique involves ball milling. In mechanical milling (MM), a suitable powder charge (typically, a blend of elemental) is placed in a high energy mill, along with a suitable milling medium. The objective of milling is to reduce the particle size and blending of particles in new phases. The different type of ball milling can be used for synthesis of nanomaterials in which balls impact upon the powder charge. The balls may roll down the surface of the chamber in a series of parallel layers or they may fall freely and impact the powder and balls beneath them. For large scale production with nano size grain, mechanical millings are more economical processes.

In our project work, we have synthesized NiFe_2O_4 by both Sol-Gel Technique and Co-precipitation Method.

2.2 Sol-Gel auto-combustion method

Sol-gel is a wet-chemical-based self-assembly process for nanomaterial formation. The sol-gel process, as the name implies, involves the evolution of networks through the formation of a colloidal suspension (sol) and gelation of the sol to form a network in a continuous liquid phase (gel). The precursors used for synthesizing the colloids generally consist of metallic ions and ligands, which are elements surrounded by various reactive species. Metal alkoxides (e.g. aluminates, titanates, borates, cuprates, etc.) and alkoxysilanes (e.g. tetramethoxysilane, TMOS and tetraethoxysilane, TEOS) are most popular because they react readily with water. In some cases, few alkoxides are immiscible in water; in that situation some homogenizing agents such as alcohols are used as mutual solvents to facilitate miscibility of these materials. In general, sol-gel formation occurs in four steps:

- i. Hydrolysis and alcoholysis
- ii. Water and alcohol condensation and polymerization of monomers to form particles
- iii. Growth of particles
- iv. Agglomeration of particles followed by the formation of networks throughout the liquid medium resulting in thick gel.

Once the gel is formed, there are several ways to convert this gel (inorganic network) to the desired solid form. Depending on the deposition and drying processes or conditions, this gel can be converted into various forms such as aerogel, xerogel, gelled spheres, nano-powders, thin film coatings, nanostructured layers, etc. Once the sol is formed, it can be converted into small gelled spheres by using various surfactants, which modify the surface tension of the dispersed

particles in the sol and prevents them from agglomeration. The gelled spheres can then be converted into powders by calcination. [3]

2.2.1 Adopted synthesis route:

A colloidal suspension of Nickel (II) nitrate hexahydrate (4.362g) and Ferric nitrate $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (12.12g) was prepared in Ethyleneglycol (30mL) taken in a beaker. In order to ensure maximum solubility, the mixture was kept on a magnetic stirrer at an rpm of 500 for 30 minutes. The sol thus obtained was kept on a Hot Plate set at 60°C . The whole system was kept in a Fume Hood since nitrate gas was bound to evolve. After 15 minutes, the sol started to boil and gelation of the sol occurred with evolution of brown fumes. After 80 minutes, auto-combustion started with the evolution of dense white fumes and continued for the next 50 minutes. Then after 5 minutes, the beaker was taken from the hot plate.

A brown residue was obtained. The residue was grinded with acetone, in a mortar for 30 minutes. A dry fine brown powder was obtained. This sample was designated S1.

2.3 Co-Precipitation method

Conventional techniques for preparation of nanoparticles and nanowires include sol–gel processing, evaporation condensation, microemulsion technique, combustion method, spray pyrolysis, hydrothermal process and template-assisted electrochemical synthesis. Generally in most of these methods the precise control over the size and its distribution is quite difficult. In order to overcome these difficulties, co-precipitation method has been used for synthesis of these nanoparticles.

The advantage of this method over the others is that the production rate of ferrite nanoparticles, its size and size distribution is relatively easy and there is no need for extra mechanical or microwave heat treatments. In this work, nickel ferrite nanoparticles (NiFe_2O_4) have also been prepared by co-precipitation technique and then they were heat treated (annealed) at different temperatures (120°C and 600°C) to synthesize magnetic nanoparticles with different

grain sizes. Various physical properties of nickel ferrite nanoparticles have been explored as a function of particle size and temperature. [7][8][9][10][15]

2.3.1 Adopted synthesis route:

Nickel ferrite (NiFe_2O_4) nanoparticles were synthesized via co-precipitation method. The synthesis procedure required 10M (800 ml) solution of sodium hydroxide (NaOH) 2M (10ml) ferric chloride (FeCl_3) and 1M (10ml) nickel chloride (NiCl_2). Using the formula for molarity, the required weight of the compounds, for the reaction to take place, was calculated. It was found that 320.001g of NaOH , 2.3769 g of $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ and 3.2442g of FeCl_3 were the required amount of compounds. 800 ml solution of NaOH was taken in a beaker and heated to 90 degree Celsius. 2 M FeCl_3 and 1 M NiCl_2 were added to this solution. This solution was vigorously stirred using the magnetic stirrer for about 1 hour at an rpm of 2200. Following this, the solution was kept for cooling and sedimenting. Next day, the supernatant was drained out and the sediment was washed with 500 mL of demineralised water. It was then, again kept for sedimentation. After the sediment was formed, some of the supernatant was again drained out and the residue was centrifuged at 5000 rpm for 10 minutes. The centrifugate was drained and the residue was washed with demineralized water. This was to reduce the pH of the sample. This centrifugation and washing with demineralized water was repeated for 5 times. By then the pH of the sample was around 7 which was confirmed with a piece of Litmus paper. Then the residue was moved to a hot air oven which was set at a temperature of 120°C .

After this, the residue was grinded with acetone, in a mortar for 30 minutes. A dry fine brown powder was obtained. This sample was designated as C1. A portion of the sample was taken and heated at 600°C in a muffle furnace for about an hour. After this, the residue was grinded with acetone, in a mortar for 30 minutes. A dry fine brown powder was obtained. This sample was designated as C2.

2.4 X-Ray Diffraction (XRD)

X-ray diffraction (XRD) is a commonly used technique to study the crystal structure of materials, including nickel ferrite. When X-rays are passed through a crystal, they interact with the electrons in the crystal lattice, causing the X-rays to diffract at specific angles, which depend on the spacing of the atoms in the crystal lattice. By analyzing the pattern of diffracted X-rays, researchers can determine the crystal structure of the material. In the case of nickel ferrite, XRD can be used to determine the arrangement of nickel, iron, and oxygen atoms in the spinel crystal structure.

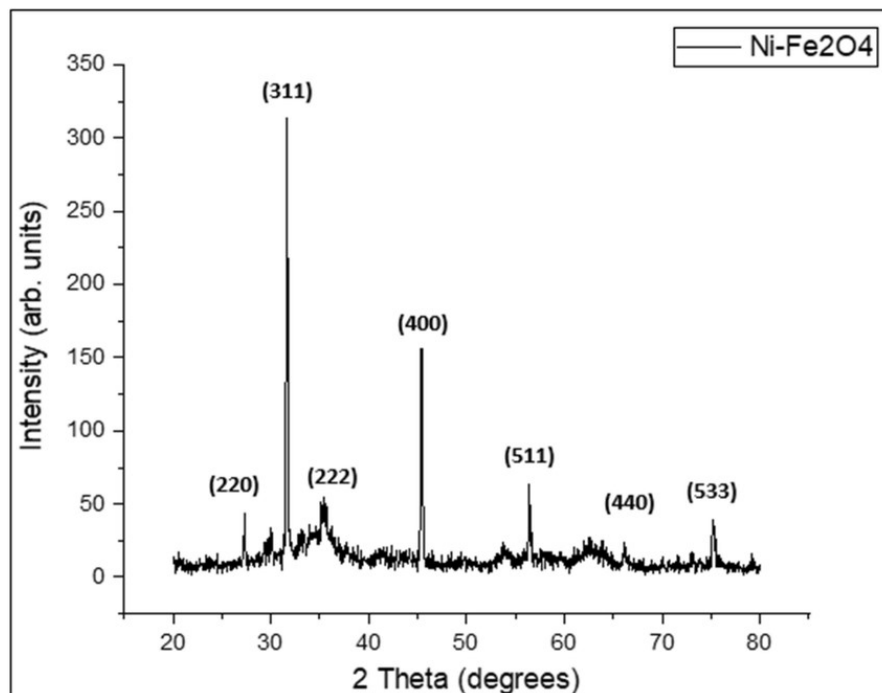


Fig 2.4.1: Reference plot of XRD of NiFe₂O₄. Picture courtesy [6]

English physicists Sir W.H. Bragg and his son Sir W.L. Bragg developed a relationship in 1913 to explain why the cleavage faces of crystals appear to reflect X-ray beams at certain angles of incidence (theta, θ). This relationship is known as Bragg's law and is given as:

$$n\lambda = 2d\sin\theta \quad (2.1)$$

The variable d is the distance between atomic layers in a crystal, λ is the wavelength of the incident X-ray beam and n is an integer and θ is the angle of incidence of X-ray. [3]

The intensity of the diffracted x-rays was measured and plotted as a function of diffraction angle 2θ . The XRD pattern of nickel ferrite typically shows several sharp diffraction peaks, corresponding to the planes of the crystal lattice that are most strongly diffracting X-rays. XRD pattern analysis is used to identify crystalline phases and orientation and to determine structural properties like lattice parameters, strain, grain size, phase composition, preferred orientation (Laue) order-disorder transformation, thermal expansion and to measure interplanar spacing (d -spacings). By comparing data with known standards in the ICDD (International Centre for Diffraction Data) file or JCPDS, which are for random orientations, we will be able to find the structure of the unknown sample.

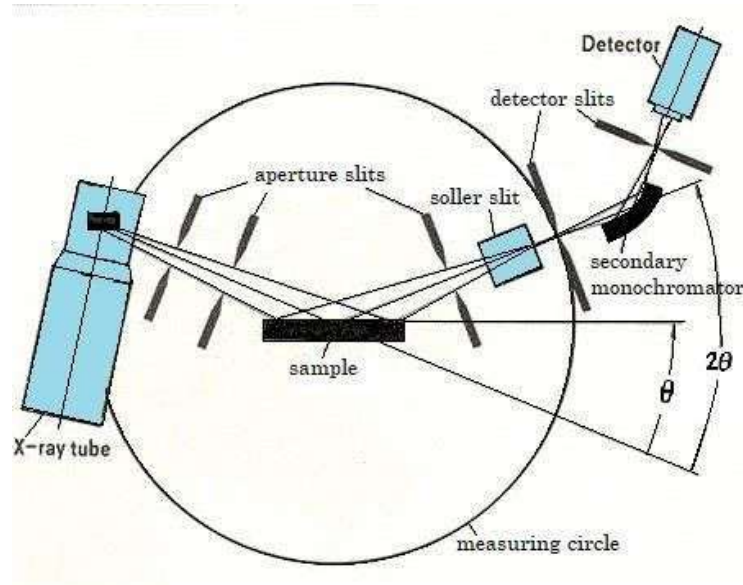


Fig 2.4.2 - Schematic diagram of X-ray diffractometer. Picture courtesy: [16]

The simplest and most widely used method for estimating crystallite size is from the Full Width at Half Maximum (FWHM) of a diffraction peak by Debye-Scherrer formula. [3]

$$D = (k\lambda/\beta \cos \theta) \quad (2.2)$$

where 'D' is the crystallite size, λ is the wavelength of the X-radiation, β is the angular width (in radians) which is equal to the (FWHM), θ is the peak position in radian and k is the Scherrer constant ($k = 0.94$). The average crystallite sizes (D) of all the samples were calculated from the XRD peak broadening of the peaks using Scherrer formula. All peaks were fitted for gaussian curve to determine the angle for maximum intensity and full width at half maximum (*FWHM or β*).

From the 2θ values of the peaks, the lattice spacing (d) values are calculated using the equation,

$$d_{hkl} = \frac{1}{\sqrt{\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}} \quad (2.3)$$

In cubic system, $a = b = c$.

The lattice parameter ' a ' was then computed by assuming cubic symmetry and combining Equations (2.1) and (2.3),

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \quad (2.4)$$

Instrumentation used for our analysis:

PANALYTICAL, Aeris Research (Source: Cu $K\alpha$)

2.5 UV-Visible Spectroscopy

UV-Visible spectroscopy (UV-Vis) measures the extinction (scatter + absorption) of light passing through a sample. Nanoparticles have unique optical properties that are sensitive to the size, shape, concentration and refractive index near

the nanoparticle surface, which makes UV-Vis spectroscopy a valuable tool for identifying, characterizing, and studying nanomaterials.

Range: In UV-Vis spectroscopy, wavelength is usually expressed in nanometers (1 nm = 10^{-9} m). The UV range normally extends from 100 to 400 nm, with the visible range from approximately 400 to 800 nm.

UV-Vis spectroscopy is a cost-effective, simple, versatile, non-destructive, analytical technique suitable for a large spectrum of organic compounds and some inorganic species. As a function of wavelength, UV-Vis spectrophotometers measure the absorption or transmission of light that passes through a medium. [11]

The band gap energy is usually determined from diffuse reflectance spectra for powdered samples. According to the theory proposed by P. Kubelka and F. Munk in 1931, the measured reflectance spectra can be transformed to the corresponding absorption spectra by applying the Kubelka-Munk function $F(R)$, given by,

$$F(R) = K/S = (1-R)^2 / 2R$$

Here, $R = R_{\text{sample}} / R_{\text{standard}}$ is the reflectance of an infinitely thick specimen, K and S are the absorption and scattering coefficients respectively. When the incident radiation is scattered in a perfectly diffuse manner, the Kubelka-Munk function is proportional to the absorption coefficient α i.e.

$$\alpha = F(R) = (1-R)^2 / 2R$$

So, by inserting Kubelka-Munk function into the Tauc relation we get,

$$(F(R)h\nu)^{1/n} = \beta(h\nu - E_g)$$

Where $n = 1/2$ for direct bandgap materials.

The above equation is similar to the equation of a straight line. So, by $(F(R)h\nu)^{1/n}$ vs. $h\nu$ plot the energy band gap of the powdered sample can be extracted easily from the X-axis intersection point. [17]

2.5.1 UV-Visible spectral analysis from reference plot:

The UV-Visible spectra of nickel ferrite sample calcined at 600°C are shown in figure 2.4.1. The band gap energies are obtained by extrapolating the linear portion of a plot of $(\alpha h\nu)^2$ versus the photon energy ($h\nu$) to the energy axis for zero absorption. The X-intercept obtained from the plot between $(\alpha h\nu)^2$ versus $h\nu$ (fig 2.4.1) gives the value of the band gap in eV. In the reference plot, the estimated band gap for nickel ferrite sample is around 2.2 eV. [16].

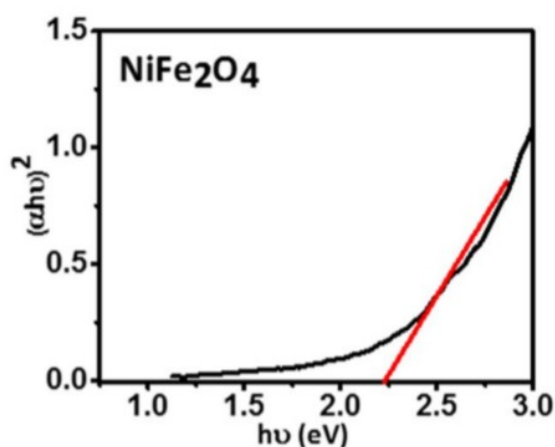


Fig 2.4.1: Reference Tauc plot from DRS data of NiFe₂O₄. Picture Courtesy: [16]

Instrumentation used for our analysis:

Integrating sphere available with UV-Vis NIR Spectrophotometer for acquiring solid state spectrum in Diffuse Reflectance mode (DRS). UV Vis NIR Spectrophotometer: Perkin Elmer Lambda 365 200 nm to 1000 nm. DRS integrating sphere 50mm, angle of incidence 8 °

2.6 Fourier Transform Infrared Spectroscopy:

Fourier Transform Infrared Spectroscopy, also known as FTIR Analysis or FTIR Spectroscopy, is an analytical technique used to identify organic, polymeric, and, in some cases, inorganic materials. The FTIR analysis method uses infrared light to scan test samples and observe chemical properties.

The FTIR instrument sends infrared radiation of about 10,000 to 100 cm^{-1} through a sample, with some radiation absorbed and some passed through. The absorbed radiation is converted into rotational and/or vibrational energy by the sample molecules. The resulting signal at the detector presents as a spectrum, typically from 4000 cm^{-1} to 400 cm^{-1} , representing a molecular fingerprint of the sample. Each molecule or chemical structure will produce a unique spectral fingerprint, making FTIR analysis a great tool for chemical identification.

FTIR spectroscopy is an established technique for quality control when evaluating industrially manufactured material, and can often serve as the first step in the material analysis process. A change in the characteristic pattern of absorption bands clearly indicates a change in the composition of the material or the presence of contamination. If problems with the product are identified by visual inspection, the origin is typically determined by FTIR microanalysis. This technique is useful for analyzing the chemical composition of smaller particles, typically 10 -50 microns, as well as larger areas on the surface. [18]

FTIR analysis is used to:

- * Identify and characterize unknown materials (e.g., films, solids, powders, or liquids)
- * Identify contamination on or in a material (e.g., particles, fibers, powders, or liquids)
- * Identify additives after extraction from a polymer matrix.
- * Identify oxidation, decomposition, or uncured monomers in failure analysis investigation

2.6.1 Analysis of FTIR from reference plot:

The FTIR spectra of annealed nickel ferrite samples recorded in the range from 400 cm^{-1} to 1000 cm^{-1} . No absorption bands are observed above 1000 cm^{-1} as a common feature of all the ferrites. All the spectra show two main absorption bands around 600 and 400 cm^{-1} . According to group theoretical calculations the spinel ferrites are known to exhibit four fundamental IR active modes in the vibration spectra, which are high frequency and low frequency bands. The FTIR spectra show strong absorption bands at 591 cm^{-1} , 561 cm^{-1} and at 481 cm^{-1} , 468 cm^{-1} for the

nickel ferrite sample annealed at 500⁰C. Similarly nickel ferrite sample annealed at 600⁰C show absorption bands at 584 cm⁻¹, 559 cm⁻¹ and at 474 cm⁻¹, 464 cm⁻¹. The band around 600 cm⁻¹, corresponds to intrinsic stretching vibration of metal at the tetrahedral site i.e, $M_{td} \leftrightarrow O$, where as the band at 400 cm⁻¹ is assigned to octahedral-metal stretching i.e. $M_{oh} \leftrightarrow O$. The IR absorption bands of solids below 1000 cm⁻¹ are usually assigned to vibration of ions in the crystal lattice. [11]

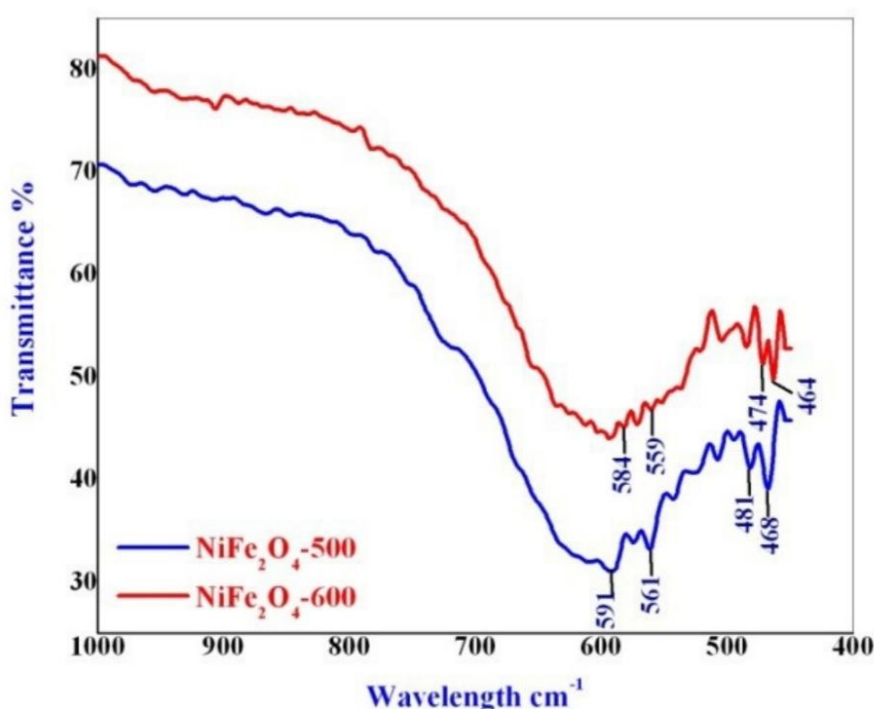


Fig 2.6.1: Reference plot of FTIR Spectrum. Picture courtesy: [11]

Instrumentation used for our analysis:

FTIR spectrometer: Thermo Nicolet iS50 4000 cm⁻¹ to 100cm⁻¹

- Resolution 0.2 cm⁻¹ , S/N Ratio 55,000:1
- ATR- Diamond crystal, Liquid Cell, Gas Cell attachments

IR library

Chapter 3

EXPERIMENTAL OBSERVATION

3.1 Introduction

The synthesized Nickel Ferrite (NiFe_2O_4) nanoparticles can be characterized using various techniques, such as X-ray diffraction (XRD), UV-Visible spectroscopy, and Fourier transform infrared spectroscopy (FTIR). XRD is used to determine the crystalline structure of the material, which can help identify the phases present and their crystallographic parameters. UV-Visible spectroscopy is used to study the optical properties of the material, such as its absorption and reflection spectra, which can provide information on the electronic structure of the material. FTIR is used to identify the functional groups present in the material and can provide information on its chemical composition and bonding.

Together, these techniques can provide a comprehensive understanding of the synthesized nickel ferrite material, its physical and chemical properties, and its potential applications.

3.2 Structural analysis using XRD

X-ray diffraction studies were carried out to investigate the structural details of the prepared NiFe_2O_4 ferrite nanoparticles. XRD pattern for the synthesized nanoparticles is given in the figure below. Figure 3.2.1 is the XRD pattern of the sample S1 and figure 3.2.2 is the XRD pattern of samples C1 and C2.

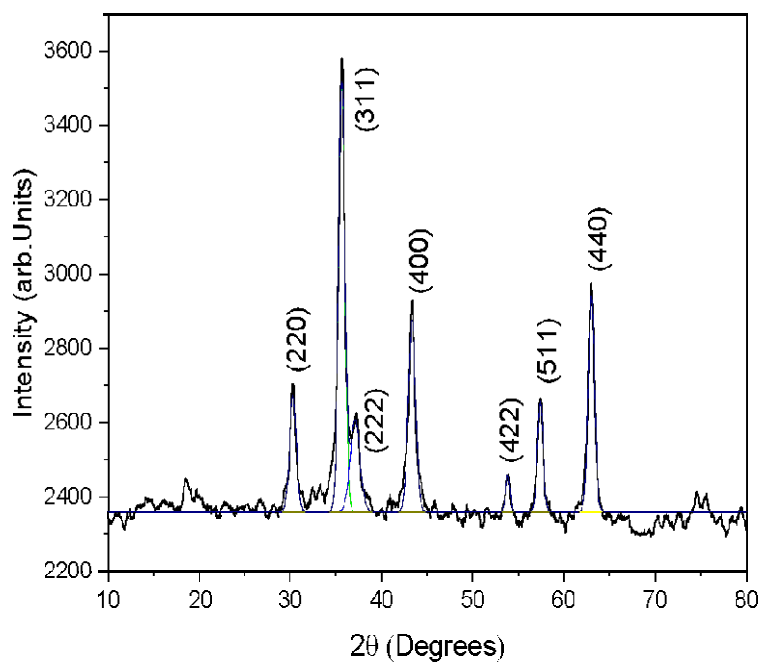


Fig. 3.2.1: XRD pattern of the sample S1

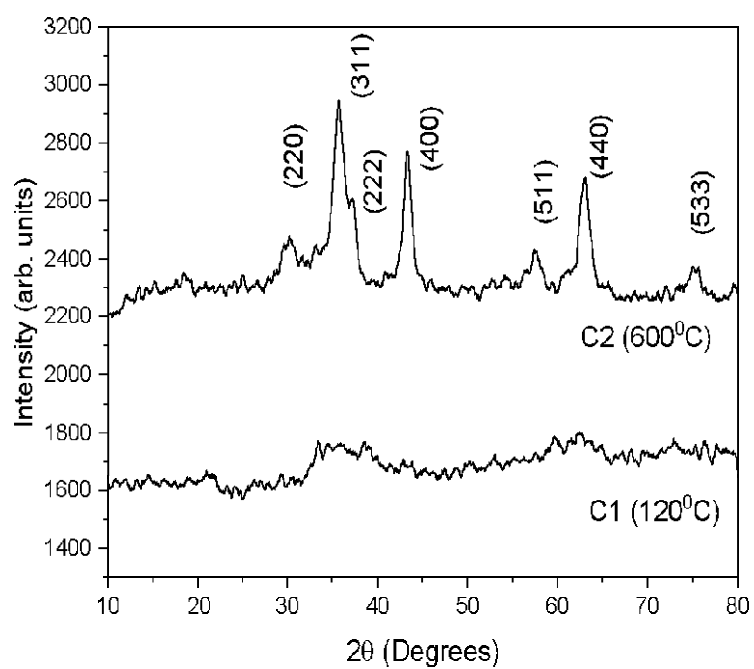


Fig. 3.2.2: XRD pattern of sample C1 and C2.

Sharp peaks appearing in the XRD pattern confirms the crystalline nature of the synthesized nickel ferrite sample. The obtained pattern was compared with standard JCPDS data for nickel ferrite (JCPDS card number 10-0325). The prominent Bragg reflections for S1 can be indexed as (220), (311), (400), (511) and (440) planes of face centered cubic spinel type structure. The prominent Bragg reflections for C2 can be indexed as (311), (400) and (440) planes of face centered cubic spinel type structure.

Using Debye-Scherrer formula, crystallite size was calculated from the broadening of the X-ray diffraction peaks. Calculated values of the crystallite size in nanometer for the prominent peaks in the XRD pattern for the sample S1 is given in Table 3.2.1.1 and for the sample C2 in Table 3.2.2.1. The average crystallite size is calculated as 10.80 nm for sample S1 and 4.63 nm for sample C2. This shows that the synthesized nickel ferrites particles are nanoparticles.

The value of the lattice parameter is calculated using interplanar distance (d) and miller indices (hkl). Table 3.2.1.2 and Table 3.2.2.2 give the lattice parameter of sample S1 and C2 respectively. The average values are calculated and found to be 8.346 Å for sample S1 and 8.325 Å for sample C2.

3.2.1 XRD Analysis of S1

2θ _B (in degrees)	Miller indices hkl	Full Width at Half Maximum (FWHM) B		Crystallite size D (nm)	Average Crystallite Size (nm)
		Degrees	Radian		
30.29	220	0.6063	0.0106	12.65	10.80
35.64	311	0.6497	0.0113	11.65	
43.34	400	0.7546	0.0132	9.79	

57.38	511	0.6318	0.0110	11.04	
62.97	440	0.7608	0.0133	8.91	

Table 3.2.1.1 Crystallite size of sample S1 calculated using Debye-Scherrer formula

2θ_B (in degrees)	Interplanar spacing d (Å)	Miller Indices			Lattice parameter a (Å)	Average value of lattice parameter (Å)
		h	k	l		
30.29	2.947	2	2	0	8.337	8.346
35.64	2.516	3	1	1	8.358	
43.34	2.085	4	0	0	8.352	
57.38	1.603	5	1	1	8.338	
62.97	1.474	4	4	0	8.344	

Table 3.2.1.2 Lattice parameter of sample S1

3.2.2 XRD Analysis of C2

$2\theta_B$ (in degrees)	Miller indices hkl	Full Width at Half Maximum (FWHM) B		Crystallite size D (nm)	Average Crystallite Size (nm)
		Degrees	Radian		
35.85	311	2.3	0.0401	3.29	4.63
43.33	400	1.178	0.0205	6.27	
63	440	1.567	0.0273	4.33	

Table 3.2.2.1 Crystallite size of sample C2 calculation using Debye-Scherrer formula

$2\theta_B$ (in degrees)	Interplanar spacing d (Å)	Miller Indices			Lattice parameter a (Å)	Average value of lattice parameter (Å)
		h	K	l		
35.85	2.5018	3	1	1	8.2977	8.3249
43.33	2.0850	4	0	0	8.3406	
63	1.4736	4	4	0	8.3364	

Table 3.2.2.2 Lattice parameter of sample C2

3.3 UV-Visible Spectroscopy Analysis

Band gap of the prepared NiFe_2O_4 was found using UV-diffuse reflectance data from UV visible spectrophotometer. UV-visible analysis of the sample was performed from 200-1000 nm. The observed reflectance spectrum for sample S1 and C2 are shown in fig 3.3.1 and fig 3.3.2 respectively.

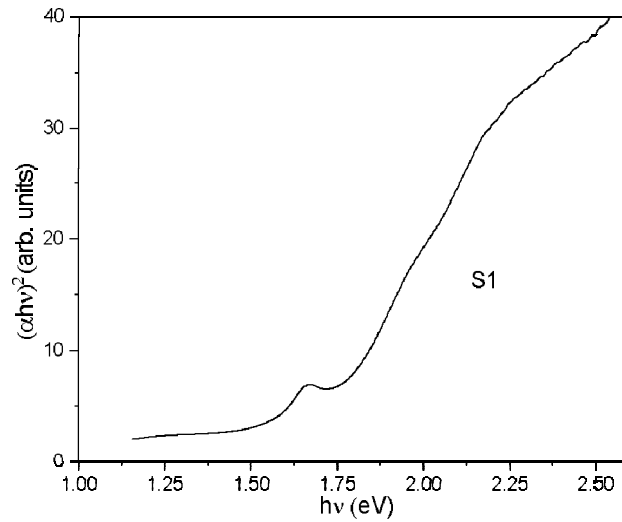


Fig 3.3.1: UV-visible reflectance spectrum of NiFe_2O_4 sample S1

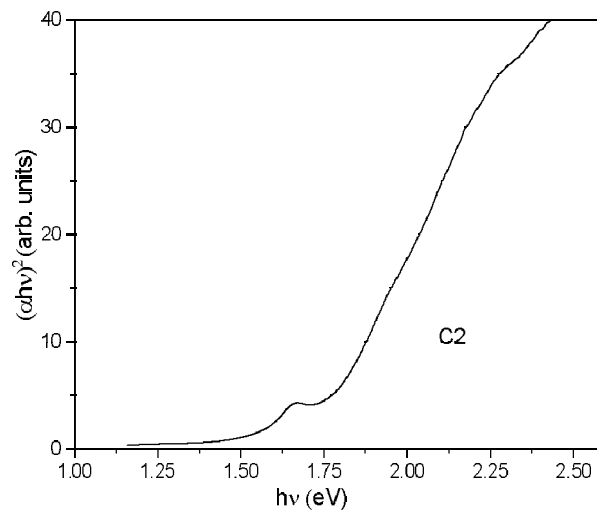


Fig 3.3.2: UV-visible reflectance spectrum of NiFe_2O_4 sample C2

The diffuse reflectance data is used to determine the band gap energy by the Kubelka-Munk function. The optical band gap is determined using the Tauc relation for direct band gap materials.

$$(\alpha h\nu)^2 = h\nu - E_g$$

Where,

$$\alpha = F(R) = (1-R)^2/2R$$

R is reflectance, and F(R) is the Kubelka-Munk function.

The X-intercept obtained from the straight line fit of the plot between $(\alpha h\nu)^2$ versus $h\nu$ gives the value of the band gap in eV. [17]

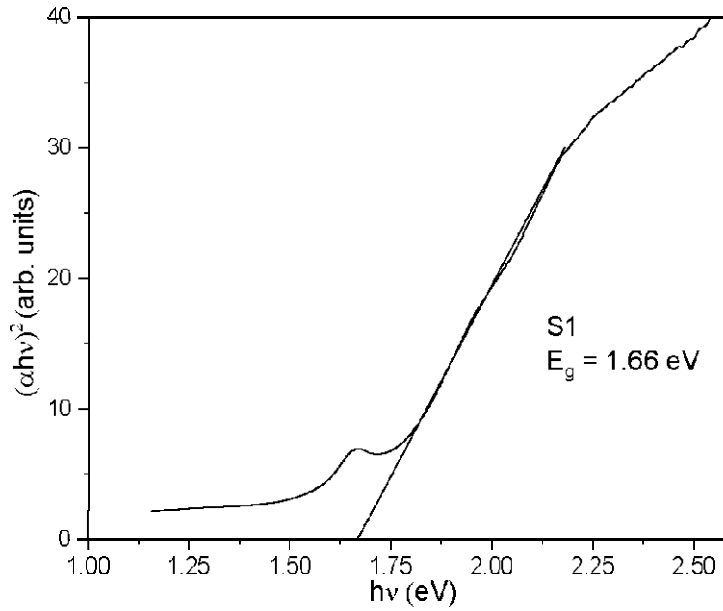


Fig :3.3.3 Tauc plot from diffuse reflectance data of NiFe₂O₄ sample S1

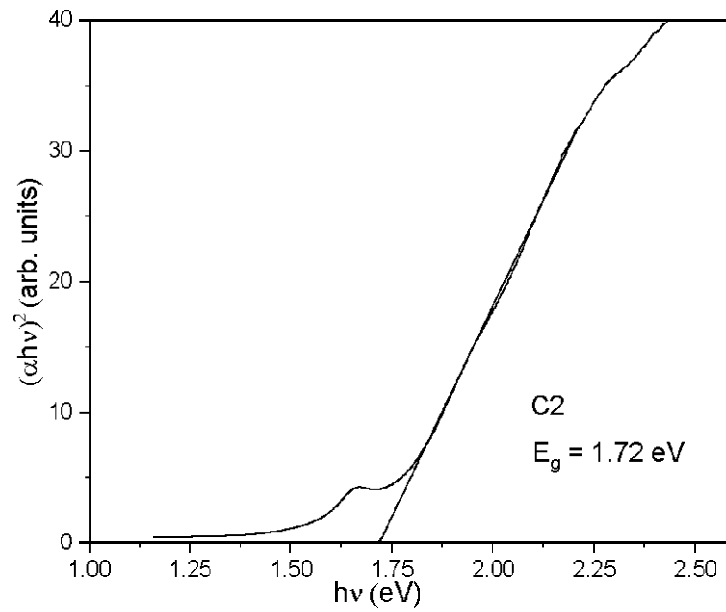


Fig :3.3.4 Tauc plot from diffuse reflectance data of NiFe₂O₄ sample C2

From X-intercept of the Tauc plot, the band gap of the prepared NiFe₂O₄ samples were found to be 1.66 eV for the sample S1 and 1.72 eV for the sample C2.

3.4 Optical Analysis using FTIR Spectroscopy

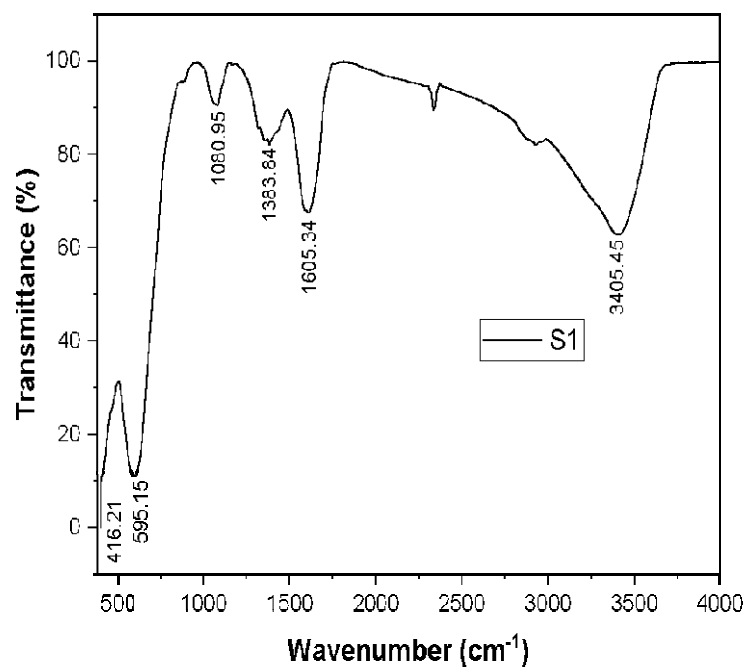


Fig 3.4.1: FTIR pattern for sample S1

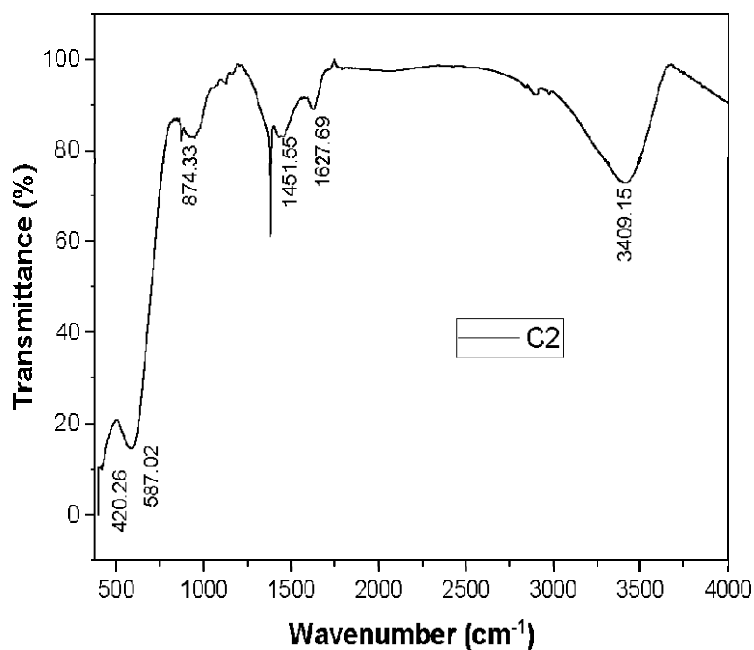


Fig 3.4.2: FTIR pattern for sample C2

As shown in Fig 3.4.1 and 3.4.2, the formation of spinel NiFe_2O_4 structure in the synthesized nickel ferrite sample is further supported by the FT-IR spectrum.

- The bands between 3400 and 3410 cm^{-1} could be attributed to the O–H stretching vibration of H_2O absorbed by the sample and the surface O–H or EG.
- The bands with the peaks at 416.21 , 595.15 cm^{-1} in Fig. 3.4.1 are assigned to the deformation vibration of Fe–OH groups in the sample S1.
- The bands with the peaks at 420.26 , 587.02 , 874.33 cm^{-1} in Fig. 3.4.2 are assigned to the deformation vibration of Fe–OH groups in the sample C2.
- The two main metal–oxygen bands at 595.15 cm^{-1} and 416.21 cm^{-1} in Fig. 3.4.1 are observed in the FT-IR spectrum of the synthesized NiFe_2O_4 sample S1.
- Similarly, 420.26 cm^{-1} and 587.02 cm^{-1} in Fig. 3.4.2 represents the two main metal–oxygen bands for sample C2.

These two bands are usually assigned to vibration of ions in the crystal lattices. This indicates the presence of uniformly distributed ferrite particles. [7]

Chapter 4

CONCLUSION

Nickel Ferrite (NiFe_2O_4) nanoparticles were synthesised in two different methods: Sol-gel auto-combustion method (sample: S1) and Co-precipitation method (samples: C1 and C2). The former being simple, low temperature and non-aqueous synthesis while the latter being high temperature and time-consuming synthesis. A comparative study of these samples in terms of different properties is done in this project work.

Structural Analysis using XRD shows that the synthesised NiFe_2O_4 nanoparticles are crystalline in nature. The absence of unwanted peaks confirms the formation of FCC lattice. The XRD pattern of Sample S1 shows its characteristic peaks thereby revealing the formation of cubic spinel phase. However, for the sample prepared through Co-precipitation method and sintered at 120°C (C1), the peaks obtained were feeble with low intensities whereas the sample prepared through the same method but sintered at 600°C (C2) shows much more defined peaks and have intensities that are higher compared to C1 but lower compared to S1. From this, we infer that, compared to the samples synthesized via Co-precipitation method, the sample synthesized via Sol-gel method yielded more crystalline nanoparticles. Among the samples C1 and C2, the latter which was heated at a higher temperature is observed to be more crystalline. The higher temperature has essentially enhanced the nanoparticle crystallinity by imparting additional nucleation energy [6]. Using Bragg's equation, the lattice parameter was estimated to be $8.325 \text{ \AA}$ for sample C2 and $8.346 \text{ \AA}$ for sample S1. These values are in accordance with the reported values [6]. Considering the relationship between width (FWHM) of the peaks in the XRD pattern and the size of the particle, the wider peaks of C2 indicates that the particles synthesized by Co-precipitation method at higher temperature yielded much more smaller particles compared to Sol-gel synthesis. This inference is confirmed by the estimation of

crystallite sizes (4.63 nm for sample C2 and 10.80 nm for sample S1) from Debye-Scherrer formula. The obtained values suggest that we have synthesized smaller nanoparticles in comparison with the reported values [6]. These estimations altogether confirm that the synthesized NiFe_2O_4 has dimensions in the nano-scale.

Band gap of the prepared NiFe_2O_4 samples were found from the UV-Visible diffuse reflectance data using the Kubelka-Munk function and they were estimated to be 1.66 eV for the sample S1 and 1.72 eV for the sample C2 indicating that they lie in the semi-conductor to insulator region. The reported band gap in other works is around 2.2 eV [16]. The possible reasons for the lower values obtained here, might include a minimal agglomeration of particles at the nanoscale, since a smaller band gap is an indicator of the larger particle size. Comparing the E_g of S1 and C2, it re-confirms our conclusion that co-precipitation yielded smaller particles compared to the sol-gel method.

The FTIR spectroscopic analysis was carried out to verify the vibrational modes of NiFe_2O_4 by comparing the obtained pattern with the characteristic peaks of the same from reference work. The two main metal–oxygen bands at 595.15 cm^{-1} and 416.21 cm^{-1} are observed in the FTIR spectrum of the sample S1. Similarly, 420.26 cm^{-1} and 587.02 cm^{-1} represent the two main metal-oxygen bands for sample C2. These two bands are usually assigned to vibration of ions in the crystal lattices. This indicates the presence of uniformly distributed ferrite particles. [7]

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